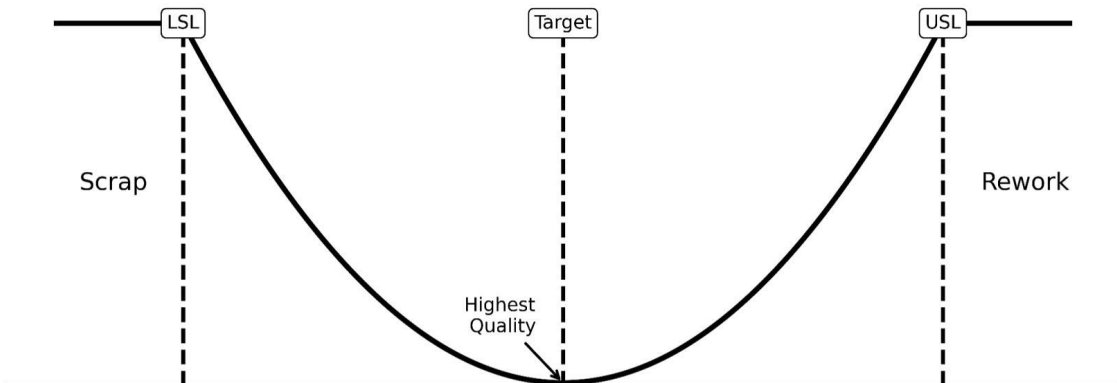


BROKEN QUALITY PRESS PRESENTS

# The definition of quality

Clarity for modern manufacturing

An essay by Jim Lehner





JIM LEHNER

# The Definition of Quality

*Clarity for modern manufacturing*

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*Second edition*

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# The definition of quality

## World-class quality

Quality has become an overused term of little substance. It has become a perverse replacement for measured characteristics of parts, products, processes, and systems that we have been trained to assume are synonymous with better. But better with respect to what? Better than the competition? Better than before? Better than expectations? Quality under the guise of better is subjective and, as a consequence, inert. This lack of inertia implies that quality cannot be improved. But what good is quality, and its associated metrics, if all they represent is stagnation? What good is a metric if it cannot be put in a context that leads to insights that result in actions and change?

Improving business and manufacturing processes requires an operational definition of quality. It requires a definition of quality that establishes where a process is while helping it get to where it needs to go. Luckily, such a definition already exists. It was introduced in September of 1960 by Dr. Genichi Taguchi. For Dr. Taguchi, world-class quality was achieved when a process or system operated

*on-target with minimum variance.*<sup>1</sup>

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<sup>1</sup> Donald J. Wheeler and David S. Chambers, *Understand Statistical Process Control* (SPC Press, 1992), xix.

What follows is a deconstruction of quality as defined by Dr. Taguchi. What follows is an evaluation of quality through the examination of its constituent parts. Such an approach aims to disembowel the notion that quality is a subjective and personal endeavor. Instead, the intent here is to establish that world-class quality is the result of thorough, repeatable, and comprehensive methods. As we will see, this definition of quality directly maps to the utility of process behavior charts as a means of facilitating improvement for both business and manufacturing processes alike. It acts as a mechanism for turning data into insights and insights into actions that result in change.

### The traditional view of quality

The traditional view of quality assumes conformance to specification is good enough. In such a view, the measured characteristics of parts, products, processes, and systems are either good or bad. As long as quality characteristics are within specification, production carries on. Shipments are made. POs are closed. Management is happy. Problems arise when specifications are not met. When measured characteristics fall outside of specification, the associated parts are deemed unacceptable. A stream of predictable and costly actions ensue. Technicians make adjustments to processes. Inspectors attempt to inspect quality into parts. Management demands that a root cause be identified, corrective actions be taken, and control plans be put in place.

This kind of quality, the traditional view of quality, is governed by a rectangular loss function like the one shown in Figure 1. The sharp cutoffs at the specification limits of this loss function indicate that all values within these limits are imbued with the same level of quality. That is to say, if it's in spec it's good enough. If, and only if, a value is greater than the Upper Specification Limit (USL) or less than the Lower Specification Limit (LSL) is the loss due to poor quality recognized. Outside of this, all values, whether they are on-target or near the spec limits, are the same. They are a homogeneous mass of uniform quality.

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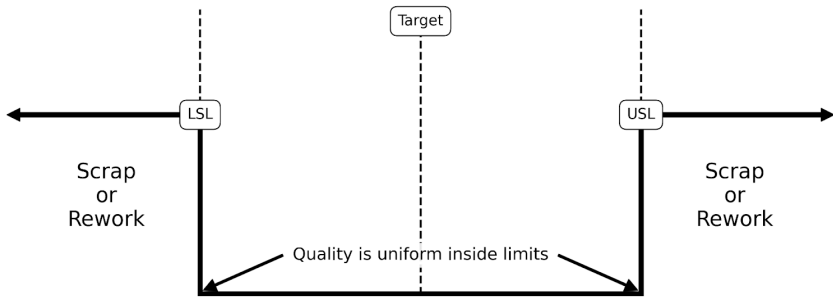


Figure 1. The traditional view of quality is governed by a rectangular loss function.

For all its good intentions and well meaning efforts, the traditional view of quality is flawed. It is built on a binary and presumptive worldview that the measured characteristics of parts, products, processes, and systems are either good or bad. Such an approach fails to accept that the process principle<sup>2</sup> always applies. That is to say, regardless of our efforts to make things the same, no two things made by the same process will ever be identical. Variation, whether it is understood or not, is always at work. In failing to understand variation, the traditional view of quality ensures defects will be produced and, by extension, world-class quality will never be achieved.

## The Taguchi loss function

In opposition to the binary methods fostered by the traditional view of quality, the Taguchi loss function evaluates quality on a continuum. This continuum, shown in Figure 2, is bound on one end by values that are on-target and on the other end by values that fall outside of specification. Values that are on-target are imbued with the highest quality while values that fall outside of specification are imbued with the lowest quality. The further a value diverges from the target, the larger the loss due to poor quality.

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<sup>2</sup> The process principle states that, whether we can measure it or not, the sequential outputs of a process are subject to some degree of difference.

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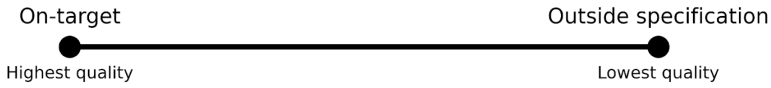


Figure 2. The quality continuum is bound on one end by values that are on-target and on the other end by values that are outside of specification. Values that are on-target are imbued with the highest quality while values outside of spec are imbued with the lowest quality.

The interstitial space of the quality continuum can be approximated, within some region close to the target, by a quadratic loss function of the form:

$$L(x) = K(x - T)^2$$

Here,  $L(x)$  is the loss function,  $K$  is a numeric constant (often expressed in terms of dollars per measurement unit squared)<sup>3</sup>,  $x$  is the value of a performance or quality characteristic, and  $T$  is the target value of the performance or quality characteristic. This quadratic loss function, when visualized as a parabola, takes the form shown in Figure 3.

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<sup>3</sup> Wheeler and Chambers, *Understanding Statistical Process Control*, 144.

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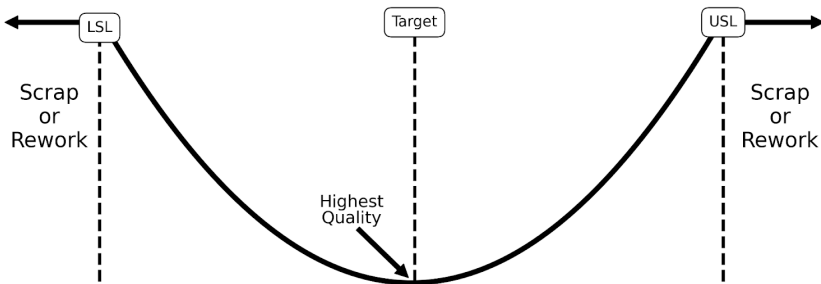


Figure 3. The Taguchi loss function is governed by quadratic loss. The further a value diverges from the target, the larger the loss due to poor quality.

The vertex of the parabola, centered at the target, is where a process will produce the highest quality. The further a value drifts from this nominal, the larger the loss incurred due to poor quality. This loss increases quadratically until the loss function reaches the specification limits. Once the specification limits are crossed, maximum loss due to poor quality is realized. In instances where rework can be performed, a portion of the loss can be recovered. In instances where parts must be scrapped, the full loss due to poor quality is realized.

The utility of the Taguchi loss function rests in its fidelity. Incremental loss, as opposed to the binary loss exhibited by the traditional view of quality, allows for sources of variation to be identified, understood, and eliminated before costs are incurred. This imbues processes and systems with an ethos of continuous improvement that embodies Dr. Taguchi's definition of world-class quality. A process that is on-target is a process that minimizes the divergence of the process average from the target. However, just because a process is on-target does not mean that it will display minimum variance. A process with minimum variance is a process that minimizes the dispersion of values around the process average. However, just because a process displays minimum variance does not mean it will be on-target. A process that is on-target with minimum variance is a process that minimizes the divergence of the process average from the target while minimizing the dispersion of values around the process average. All four of these conditions are visualized using

the histograms in Figure 4.

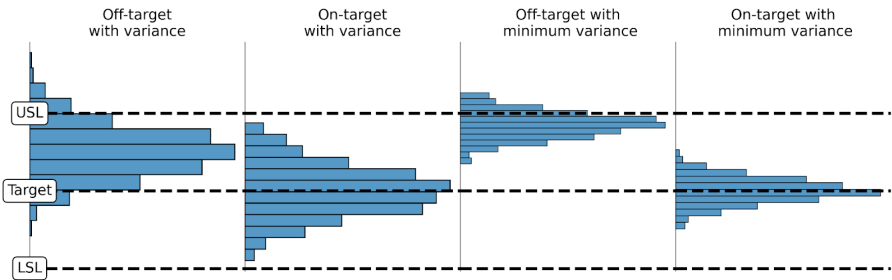


Figure 4. On-target with minimum variance is one of four operational states for a process. Processes can also operate off-target with variance, on-target with variance, or off-target with minimum.

While the use of histograms like those shown in Figure 4 are a step toward identifying processes that display the characteristics of world-class quality, they are only a step. Histograms have no mechanism for characterizing process behavior as either predictable or unpredictable. As such they have no means of informing an operator if a process displays *a reasonable degree of statistical control*.<sup>4</sup> The only tool capable of such a task is the process behavior chart. Thus, in addition to the knowledge of how to set the process average at the target, the knowledge of characterization, whether past process behavior can be described as predictable or unpredictable, is a quintessential element of world-class quality. Without this knowledge it is impossible to anticipate future process behavior based on the past. While histograms can allude to such a state, only process behavior charts can verify it.

In the pursuit of world-class quality, care should also be taken to avoid

<sup>4</sup> The state of statistical control occurs when, through using a process behavior chart (control chart), all individual values are less than the Upper Process Limit (UPL) and greater than the Lower Process Limit (LPL) on the X Chart, and all moving ranges are less than the Upper Range Limit on the mR Chart.

confusing process limits<sup>5</sup> and specification limits. Where process limits are values calculated from process data and thus represent the *voice of the process*, specification limits are requirements defined by the customer and thus represent the *voice of the customer*. Process limits bound what a process is doing, while specification limits bound what a customer is willing to pay for. To confuse the two, either process limits for specification limits or specification limits for process limits, leaves you vulnerable to making one of the two types of mistakes.

Mistake 1 occurs when common causes of routine variation are confused for assignable causes of exceptional variation. Mistake 2 occurs when assignable causes of exceptional variation are confused for common causes of routine variation. Such confusion can lead to taking actions when actions are not necessary and complacency when actions are necessary. In either case, the confusion that arises from conflating the *voice of the process* with the *voice of the customer* undermines efforts to operate processes on-target with minimum variance.

## Opposing views produce opposing results

In contrast to the traditional view of quality, the Taguchi loss function is a more realistic model of loss. Where the traditional view of quality focuses on meeting specification, the Taguchi loss function focuses on consistency. Where the traditional view of quality promotes idleness, the Taguchi loss function promotes action. Naturally, these opposing views produce opposing results.

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<sup>5</sup> Process limits are also called natural process limits or control limits. Regardless of the name, these limits are calculated using process data, the mean of the process data, a scaling factor, and the average moving range of the process data.

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| <b>Traditional view</b>      | <b>Taguchi's view</b>      |
|------------------------------|----------------------------|
| Conformance to specification | Focus on consistency       |
| Quality is binary            | Quality falls on continuum |
| Rectangular loss function    | Parabolic loss function    |
| Homogenous                   | Heterogenous               |

Figure 5. Opposing views of quality produce opposing results.

For the traditional view of quality, the focus on adherence to specification produces outcomes that maximize variation. This is the unavoidable consequence of a binary framework. When everything inside of specification is acceptable, quality has no fidelity. Regardless of whether a value is on-target or near the specification limits, all values that meet specification are the same.

On the other hand, the Taguchi loss function, with its focus on consistency, produces outcomes that minimize variation. This is the unavoidable consequence of a high fidelity, continuum based framework. By replacing the rectangular loss function that governs the traditional view of quality with the Taguchi loss function, Dr. Taguchi admonished that loss due to poor quality is incremental. Incremental loss, as opposed to the sudden loss exhibited by the traditional view of quality, can be tracked and monitored. This allows individuals, teams, and organizations to preemptively respond to issues before loss due to poor quality is realized. Moreover, in instances where loss due to poor quality has occurred, a postmortem of where things went wrong can be performed and efforts can be made to stop them from happening again in the future.

## Summary

To adopt and maintain the traditional view of quality is to adopt and maintain methods that, for the better part of 200 years, have yet to produce results. Processes and systems that are managed by the traditional view reliably and repeatedly fall short. They see conformance to specification as the finish line and trade world-class quality for good enough. Operating under the false

pretext that good enough is, well, good enough, the traditional view of quality ensures that defects will inevitably be produced.

Conversely, to adopt and maintain the methods that produce world-class quality, is to adopt and maintain methods that relentlessly extort processes and systems into doing more than it was thought they could achieve. Processes and systems that are governed by the Taguchi loss function reliably and repeatedly outperform not just the competition but themselves. They are bound to the iterative and “dynamic scientific process of acquiring knowledge”<sup>6</sup> called the Plan-Do-Study-Act (PDSA) cycle, that incessantly promotes improvement.

Operating under the axiom that no two things are alike, the Taguchi loss function ensures that time, attention, and resources are directed to areas that need it most. It ensures a dedicated approach to continual improvement and, as a consequence, it ensures world-class quality can be achieved. However, world-class quality can only be realized when management wholeheartedly embodies its underlying and driving ethos. This ethos was quietly introduced in September of 1960 by Dr. Genichi Taguchi. This ethos defines world-class quality as *on-target with minimum variance*.

## Notes

- World-class quality was defined by Dr. Genichi Taguchi in September of 1960 as *on-target with minimum variance*.
- There are two views of quality: the traditional view and the Taguchi loss function.
- The binary traditional view of quality deems parts, products, processes, services, and systems as either good or bad.
- The Taguchi loss function views quality on a continuum where parts with the highest quality have measured characteristics equal to the target and parts with the lowest quality have measured characteristics that fall

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<sup>6</sup> Walter A. Shewhart, *Statistical Method from the Viewpoint of Quality* (Dover Publications, 1986), 45.

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outside of specification limits.

- To achieve world-class quality we must use the process behavior chart which reflects the high fidelity continuum based framework of the Taguchi loss function.